



Measurement and Calculation Methods on Absorption and Scattering Properties of Turbid Food in Vis/NIR Range

Dandan Xie¹ · Wenchuan Guo^{1,2,3}

Received: 21 September 2019 / Accepted: 2 January 2020
© Springer Science+Business Media, LLC, part of Springer Nature 2020

Abstract

In terms of optical properties, most of the foods are turbid. The ability of light to penetrate a turbid material mainly depends on its absorption and scattering properties. Understanding the absorption and scattering properties of food is not only helpful to understand their chemical and physical properties, but also useful to optimize and develop effective devices for quality determination of food. To provide information for further studies on absorption and scattering properties of turbid food, the measurement techniques, such as integrating sphere, spatially resolved, frequency-domain, and time-resolved spectroscopes, and calculation algorithms, such as inverse adding double, Monte Carlo, and diffuse approximation, on obtaining absorption and scattering properties over the visible and near-infrared wavelength range were reviewed. The characteristics of these methods as well as the future trend on studying absorption and scattering properties of turbid food were also presented.

Keywords Optical property · Absorption property · Scattering property · Turbid material · Measurement techniques · Calculation algorithms

Abbreviations

AD	Adding double
DA	Diffuse approximation
DIS	Double integrating sphere
DTOF	Distribution of time-of-flight
FD	Frequency domain
HI-SRS	Hyperspectral imaging-based spatially resolved spectroscopy
IAD	Inverse adding double
IS	Integrating sphere
MC	Monte Carlo
MI-SRS	Monochromatic imaging-based spatially resolved spectroscopy
NIR	Near-infrared
RTE	Radiative transfer equation

SIS	Single integrating sphere
SNR	Signal-noise ratio
SRS	Spatially resolved spectroscopy
SFDI	Spatial frequency-domain imaging
SFDI-SRS	Spatial frequency-domain imaging-based spatially resolved spectroscopy
SR	Spatially resolved
TR	Time-resolved
TRS	Time-resolved spectroscopy
Vis/NIR	Visible and near-infrared

Introduction

Food, which provide nutrients, energy, and sensory pleasure for humans, are produced and consumed in a wide varieties of forms. In terms of optical properties, food and agricultural products could be divided into two types, i.e., transparent (including semitransparent or translucent, such as beverages, cooking oils, fruit juices, jelly) and turbid (also known as opaque, e.g., meats, grains, nuts, dairy, fruits, and vegetables). In transparent food, light passes through mediums straightly without being scattered or changing direction. While in turbid food, which are made up of complex biological components, such as water, fat, protein, lactose, citric acid, and inorganic

✉ Wenchuan Guo
guowenchuan69@126.com; wencg915@sina.com

¹ College of Mechanical and Electronic Engineering, Northwest A&F University, Yangling 712100, Shaanxi, China

² Key Laboratory of Agricultural Internet of Things, Ministry of Agriculture and Rural Affairs, Yangling 712100, Shaanxi, China

³ Shaanxi Key Laboratory of Agricultural Information Perception and Intelligent Service, Yangling 712100, Shaanxi, China

compounds, light would change direction and undergo multi-scattering at each scattering event until it reemerges from the materials or is absorbed. Actually, most foods are turbid, which means that multi-scattering occurs in them. The ability of light to penetrate a turbid material mainly depends on its absorption and scattering properties, i.e., absorption and scattering are main interactions of light with turbid biological tissues (Lu 2016). Light absorption is mainly influenced by the chemical compositions of food, such as sugar content, fat, and protein, while scattering is related to the structural and physical properties, such as cell size and density. Each food has its unique chemical and structural characteristics, which in turn determine its unique absorption and scattering properties, which provides important information about the propagation of light in food.

The visible and near-infrared (Vis/NIR) spectroscopy has been proved having great potential in food detection on composition, safety, and quality and has been applied in some industries. However, the integrated influence of absorption and scattering events on obtained Vis/NIR spectra of tissue makes it hard to illustrate the chemical and physical properties from the meaningful information presented in obtained spectra, since a change in obtained spectra may reflect a change in absorption, scattering, or both. Quantification on absorption and scattering properties is helpful to understand the chemical and physical properties of observed biological tissue, to know underlying mechanism that permit quality detection and light propagation in biological tissue, and to develop effective devices for quality determination of food (Hu et al. 2018; Palmer and Ramanujam 2006).

The absorption and scattering properties can be obtained by using proper techniques and algorithms. The techniques used to measure these properties over the Vis/NIR wavelength range mainly including integrating sphere (IS) (Jacquez and Kuppenheim 1955), spatially resolved (SR) (Verhoelst et al. 2011), frequency-domain (FD) (Carp et al. 2017), and time-resolved (TR) (Patterson et al. 1989) spectrometers. These techniques allow the separation of absorption and scattering properties, while the conventional Vis/NIR point spectroscopic measurement cannot achieve this. The algorithms applied in calculating absorption and scattering properties usually include inverse adding double (IAD), Monte Carlo (MC), and diffuse approximation (DA). Combined with DA, the absorption property and scattering property can be estimated by means of TR, FD, and SR through measuring the diffuse reflectance of the material. Similarly, the absorption and scattering properties can also be obtained by using IS with IAD.

The objective of the review was to provide a summary on measurement systems and calculation algorithms for absorption and scattering properties of turbid food in Vis/NIR range. The problems as well as the development of this study in the future were also presented.

Absorption and Scattering Properties and Main Optical Parameters of Food

Absorption and Scattering Properties

Absorption Property

As light travel through the tissue, the intensity of light gradually weakens, which is called light absorption. The absorption ability is expressed by absorption coefficient, μ_a , which is defined as the probability that a photo is absorbed after being propagated per unit length. The absorption of light follows the Lambert-Beer law. When there is only absorption phenomenon, the absorbed light in the tissue can be expressed by Eq. (1).

$$I = I_0 e^{(-\mu_a d)} \quad (1)$$

where I is the intensity of light after it passes through the medium, I_0 is the intensity of incident light, and d is the thickness of medium in millimeters or centimeters.

Owing to the molecule transition from ground state to high energy level caused by anharmonic vibration, the absorption of light is related to the chemical composition of mediums. Pigments, such as carotenoid, anthocyanin, and chlorophyll, are main light-absorbing groups in visible light range. While in NIR range, the hydrogen bonds, e.g., C–H, N–H, O–H, F–H, which are ample in food, have strong interactions with light. Carbohydrate and water are main NIR light-absorbing groups in food. Therefore, the information of the substance, such as the compositions and their contents, can be derived from the spectrum of absorption, which is the basis of Vis/NIR spectroscopy as a nondestructive measurement technique.

Scattering Property

Scattering of photons is a physical procedure greatly decided by wave characteristics of photons and structures of materials that they travel through, including particle size or shape and material's density. Light scattering has two forms, i.e., elastic and inelastic scattering. In the case of the former, no net energy exchange occurs to the photons, or the frequency of photons does not change after they have interacted with particles. While in the latter, after interplaying with atoms or molecules in the medium, those photons lose or gain energy and subsequently change the frequency.

In an optically homogeneous scattering materials, light scattering can be expressed by the Lambert-Beer law, i.e.,

$$I = I_0 e^{(-\mu_s d)} \quad (2)$$

where μ_s is the scattering coefficient, describing the probability that a photo is scattered after being propagated per unit length. Usually, the scattering property is described by μ_s .

Be scattered by the particle, the photo is deviated from the original direction, and the deflection angle is θ , shown in Fig. 1. The distribution probability of θ can be expressed by scattering phase function, designated as $p(\cos\theta)$. Currently, the most commonly used method for calculating $p(\cos\theta)$ is the function of Henyey-Greenstein (Tuchin 2007), that is

$$p(\cos\theta) = \frac{1-g^2}{2(1+g^2-2g\cos\theta)^{\frac{3}{2}}} \quad (3)$$

where anisotropy factor g is the cosine expectation of θ and is calculated by Eq. (4)

$$g = \langle \cos\theta \rangle = \int_0^\pi p(\theta) \cos\theta \cdot 2\pi \sin\theta d\theta \quad (4)$$

The values of g are between -1 and 1 . $g = -1$, 0 , and 1 represent total backward scattering, isotropic scattering, and total forward scattering, respectively. Owing to the inverse of $g = 0$ is not necessarily true, $g = 0$ only represents the integral over -1 and 1 (Welch and Gemert 2011). For most turbid mediums, the value of g ranges from 0.70 to 0.99 , indicating that forward scattering is dominant.

Since light scattering is associated with physical and structural properties (e.g., cell structure) of medium, it is often used to predict hardness of the medium.

Optical Parameters

Refractive Index

When a beam of incident light obliquely enters one medium from another, the direction of propagation will change, that is light refraction. The refractive index n , a basic physic parameter, can be utilized to describe the propagation of light in medium. It is defined as the ratio of light propagation velocity in vacuum to a medium.

$$n = \frac{c}{v} \quad (5)$$

where c and v are velocities of light in vacuum and in a medium, respectively.

According to refraction law or Snell's law, the change of light direction is determined by refractive index as it passes from one medium to another, that is,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \quad (6)$$

where n_1 and n_2 are refractive indices of the two media, and θ_1 and θ_2 are incident angle and refraction angle, respectively.

Other Optical Parameters

Other optical parameters derived from above-described basic parameters are also commonly used, including total attenuation coefficient (μ_t), reduced scattering coefficient (μ_s'), mean free path (mfp), effective attenuation coefficient (μ_{eff}), effective depth of penetration (δ), and albedo (α).

μ_t indicates the probability of light attenuation per unit length, and μ_s' reflects combined effect of μ_s and g . mfp indicates the path that photons have traveled before being absorbed and scattered. μ_{eff} indicates the ability of light to penetrate medium. δ equals to the reciprocal of μ_{eff} , and α indicates the ability to generate scattering, and calculated by μ_a and μ_s . These parameters can be calculated by the following equations:

$$\mu_t = \mu_a + \mu_s \quad (7)$$

$$\mu_s' = \mu_s(1-g) \quad (8)$$

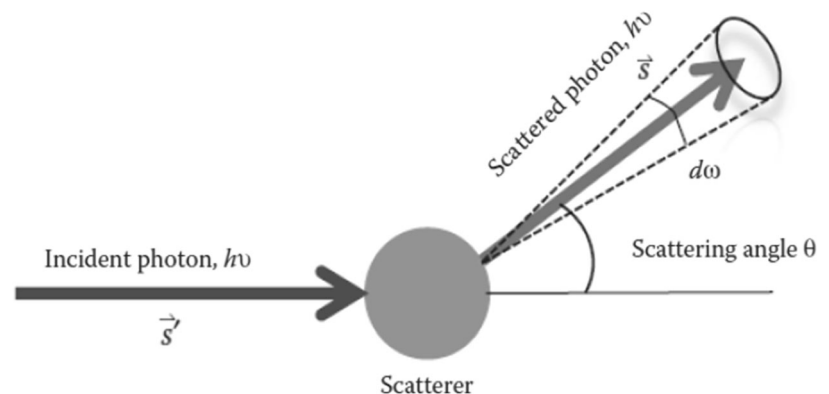
$$mfp = \frac{1}{\mu_t} = \frac{1}{\mu_a + \mu_s} \quad (9)$$

$$\mu_{eff} = [3\mu_a(\mu_a + \mu_s')]^{\frac{1}{2}} \quad (10)$$

$$\delta = \frac{1}{\mu_{eff}} \quad (11)$$

$$\alpha = \frac{\mu_s}{\mu_a + \mu_s} \quad (12)$$

Fig. 1 A scattering event occurs when an incident photon with energy $h\nu$ hits the scatter along the direction $\vec{s}'$, and it is then scattered into the direction $\vec{s}$ (Lu 2016)



Measurement Techniques on Absorption and Scattering Properties

The absorption and scattering properties of turbid food can be measured using either direct or indirect methods (Tuchin 2007). At present, the techniques used to measure these properties mainly includes IS, SR, FD, and TR techniques. Contrasted with conventional Vis/NIR point spectroscopic measurement, these techniques allow the properties of absorption and scattering be separated. In this section, these measurement systems were reviewed.

Integrating Sphere Technique

IS is a hollow sphere coated with diffuse reflection materials whose surface reflectivity is up to 98% on its inner surface. Due to the interior surface of integrating sphere can receive and reflect all incident light and **it does not absorb any light**, the brightness observed from any viewing angle is same, whatever the incident light irradiated to the interior surface are distributed.

The IS system used in measuring absorption and scattering properties includes single integrating sphere (SIS) and double integrating sphere (DIS), which are characterized by the number, diameter, ports, baffle, and wall reflectance (typically 98%) of sphere. When a beam of incident light enters the inside of an integrating sphere from its entrance port, it is reflected back and forth inside the integrating sphere. Finally, uniformly diffused light is obtained at the exit port. Moreover, arbitrary change on the incident angle and spatial distribution of light has no effect on the output light. The ratio of the exited light intensity to the incident light intensity is about the ratio of the area of light output aperture to surface area of the sphere.

Single Integrating Sphere

A SIS system includes an integrating sphere, a spectrometer, a light source, and a fiber collimator. Figure 2 is the schematic diagram of a SIS system and the experimental setup for measuring the total diffuse transmittance T_T , total diffuse reflectance T_R , and the collimated transmittance C_T . In this system, two ports oppositely located on the equator of the sphere are utilized as entrance and exit, and the port plug on the north pole is connected to the spectrometer using an optical fiber. The spectrometer is used as the detector to obtain spectral information.

When measuring the spectra of T_T , the exit port is blocked by a port cover, and the sample is placed in front of the entrance port (Fig. 2a). While, for the measurement of T_R , the sample is placed at the exit port of the sphere, and the entrance port is open (Fig. 2b). In this mode, the collimator is placed at a suitable position to obtain the same size of illumination spot

as that in T_T measurements. To measure C_T , the center of the detecting optic fiber is aligned with the light beam. Between the collimator and the sample, an iris is placed to constrain the position and intensity of incident beam. Another iris, placed between the sample and the detecting optic fiber, is applied to avoid the ambient light entering the detector. The detailed approach on measuring the collimated transmittance could be found elsewhere (Prahl 2011).

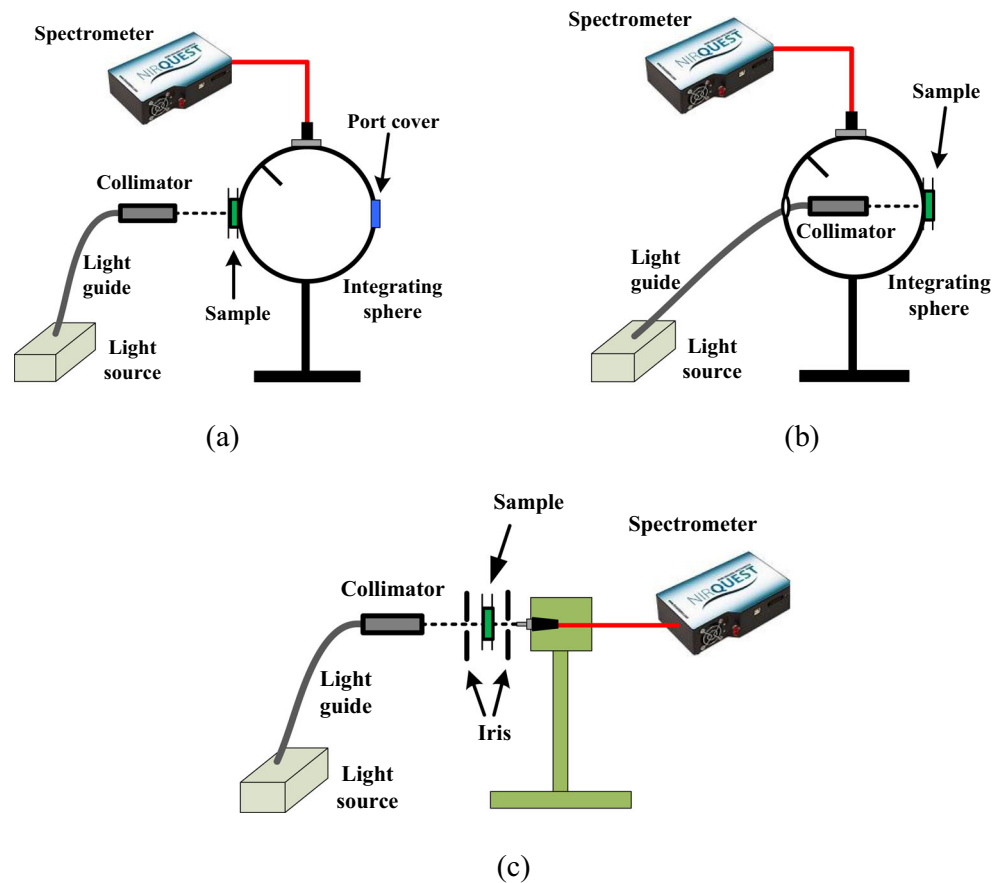
Based on the obtained T_T , T_R , and C_T , the absorption and scattering properties or optical parameters, such as μ_a and μ_s' , can be calculated by applying IAD method, which was described in the “Inverse Adding Doubling” section.

Double Integrating Sphere

Double integrating sphere system, as its name implies, has two spheres. The sample is placed between the two spheres. As shown in Fig. 3, the unscattered collimated transmitted light is allowed to leave the system. The two spheres could have different geometries and surface reflectance, while the area of the sample ports should be same. The reflectance and transmittance factors and sample shape are not related to the side of sample where the light is incident. Once light irradiates the sample in reflectance sphere (that is the left sphere), a portion of light passes through the sample to transmittance sphere (the right sphere). Within the transmittance sphere, a portion of light will irradiate the back of the sample, then a part of them are transmitted back into the reflectance sphere again. Therefore, the signal of light is increased in the reflectance sphere. Afterwards, the remaining light in the reflectance sphere is (re)transmitted to the transmittance sphere. Similarly, the signal is increased in the transmittance sphere. The transmission procedure of an ever-decreasing part of light forth and back through the sample mentioned above continues until the light in the sphere has been completely absorbed or disappears. As SIS, the absorption and scattering properties can be obtained by combining with IAD method.

In DIS, the transmittance and reflectance of light can be measured simultaneously, but they just can be measured at the entrance and exit ports, respectively, in SIS. However, bigger errors are existed in DIS than in SIS, causing uncertainties of measurement. Up to now, the system of integrating sphere has been applied to estimate optical properties of fruit, such as kiwifruit (Liu et al. 2019a; Fang et al. 2016), apple (Wei et al. 2020; Saeys et al. 2008; Zhang et al. 2017), blueberry (Zhang et al. 2019) and pear (He et al. 2016), vegetable, such as onion (Wang and Li 2013) and potato (López-Maestresalas et al. 2016), meat (Zamora-Rojas et al. 2013), and milk (Aernouts et al. 2015a, 2015b). IS system is suitable for initial research on optical properties of tissues since it has higher accuracy than other measurement techniques.

Fig. 2 Schematic of the single integrating sphere system and the experimental setup for measuring the total diffuse transmittance T_T (a), total diffuse reflectance T_R (b), and collimated transmittance C_T (c) (Liu et al. 2019b)



Spatially Resolved Spectroscopy

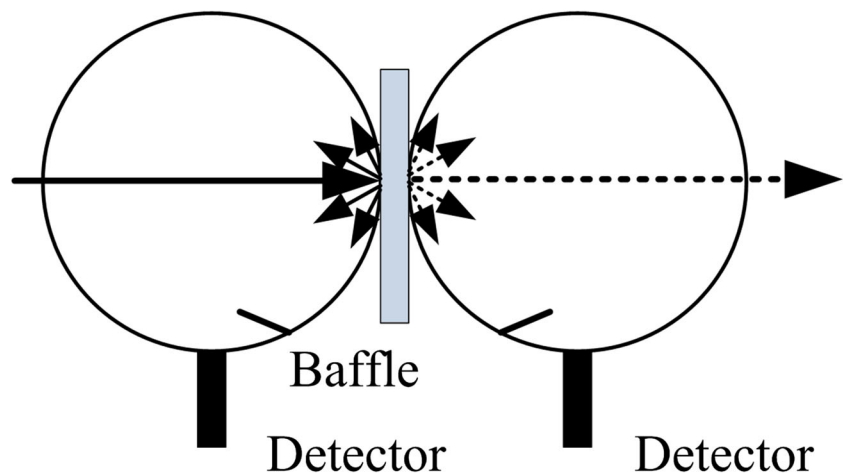
Figure 4 shows the principle of estimating absorption and scattering properties by using spatially resolved spectroscopy (SRS). As shown in Fig. 4, the diffusely reflected light from the sample surface is measured at different irradiated source-detector distances. Based on different principles of SRS, there are different measurement devices, mainly including fiber-optic probe, monochromatic imaging-based spatially resolved spectroscopy (MI-SRS), hyperspectral imaging-based

spatially resolved spectroscopy (HI-SRS), and spatial frequency-domain imaging-based spatially resolved spectroscopy (SFDI-SRS).

Fiber-Optic Probe

Figure 5 is the schematic illustration of a fiber-optic probe setup, mainly consisting of a light source, a fiber-optic probe, a spectrograph, a camera, a data acquisition device, and a computer. The light source, such as halogen lamp, delivers

Fig. 3 Arrangement of double integrating sphere (Pickering et al. 1993)



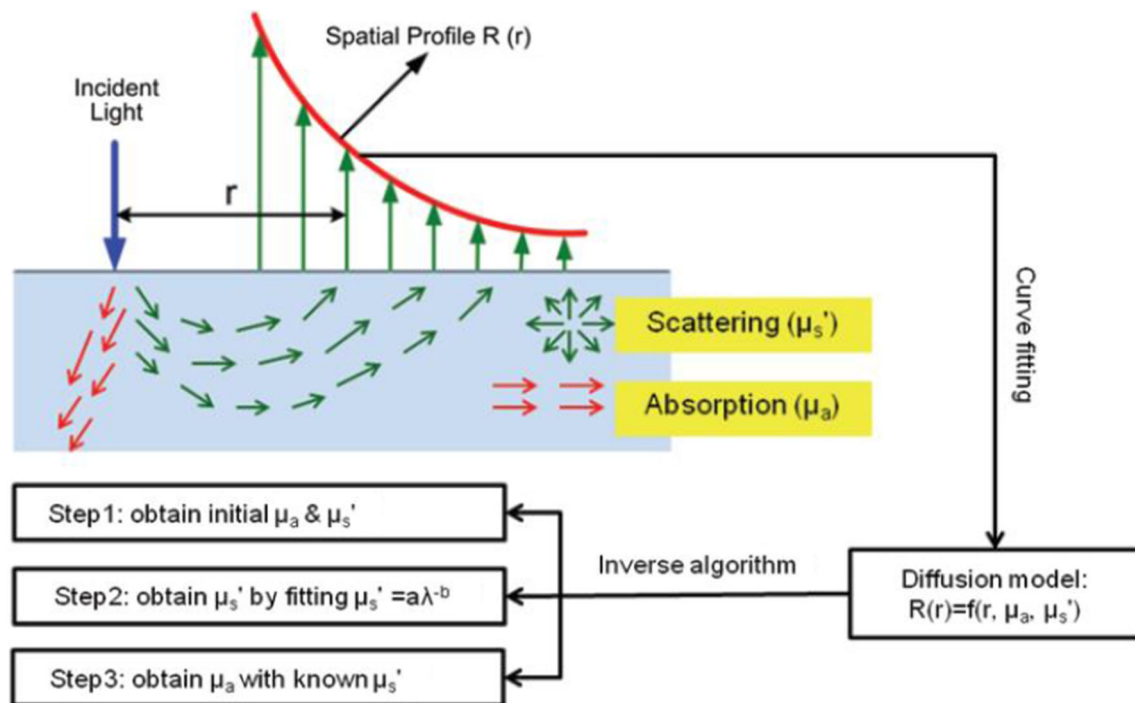


Fig. 4 Principle of estimating optical properties using spatially resolved spectroscopy (Cen et al. 2012)

light to irradiate the sample through an illumination fiber. Several detection fibers are set at different center-to-center distances to collect diffuse reflected photons. The collected diffuse reflected photons are introduced to the entrance slit of a spectrograph through the detection fibers. Then, the light from each of these fibers is split into spectral components, and the split corresponding photos are projected onto a CCD camera by spectroscopy. Finally, the signal from the camera is

transmitted to a computer through a data acquisition device, such as a data acquisition card.

The fiber-optic probe method needs good contact between sample and detecting probe. Moreover, long measurement time is needed, and fluctuation in output of light source is existed in this method. All these limit the applicability of this method for scientific research (Cen et al. 2016), especially for some solid food products. However, the fiber-optic probe

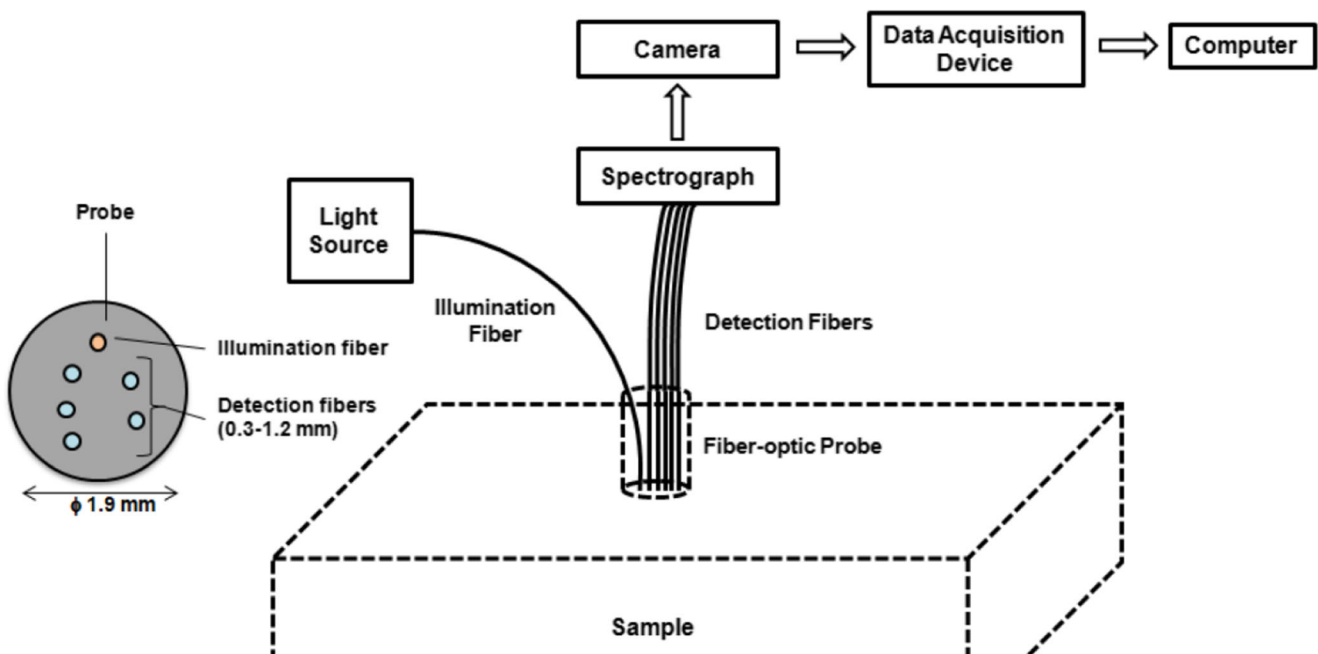


Fig. 5 Schematic illustration of the fiber-optic probe setup (Watte et al. 2013)

method has a higher signal-noise ratio (SNR) than hyperspectral cameras applied in other systems of SRS. Recently, the SRS technique based on fiber-optic probe has been applied in assessing the quality of apple (Nghia Nguyen Do et al. 2013; Nghia Nguyen Do et al. 2014a), characterizing the texture and microstructure of aerated sugar containing food and dried apple slices (Herremans et al. 2013; Nghia Nguyen Do et al. 2014b) and estimating the optical properties of beef muscle (Xia et al. 2007, 2008).

Monochromatic Imaging-Based Spatially Resolved Spectroscopy

Figure 6 shows a diagram of a MI-SRS system. Laser diode at a single or several wavelengths is usually used as the light source to emit a beam to the surface of a turbid medium at a specific wavelength. The CCD camera is used to obtain the spatially resolved diffuse reflectance, allowing obtaining a set of reflectance values simultaneously from the surface of a turbid medium. A temperature controller and a current controller are used to make the system be in a steady state. The advantage of the system is achieving noncontact measurement, especially for food whose sanitation and safety requirements are important. However, the absorption and scattering properties just can be measured at a single or several wavelengths. Up to now, the MI-SRS system has been applied to detect the quality of fruit, such as citrus (Lorente et al. 2013) and banana (Adebayo et al. 2016).

Hyperspectral Imaging-Based Spatially Resolved Spectroscopy

Combined spectroscopy and imaging methods together, hyperspectral imaging technique could obtain both spectral and spatial information simultaneously. Therefore, the technique is ideally appropriate for obtaining spatially resolved diffuse reflectance profiles over a broad wavelength range. Figure 7 shows the schematic diagram of a HI-SRS system. A

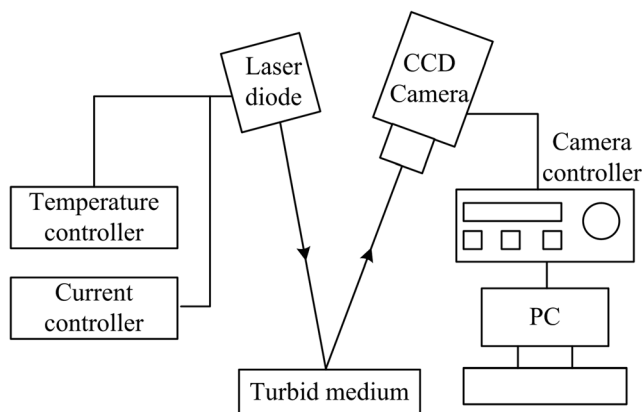


Fig. 6 Diagram of a monochromatic imaging-based spatially resolved spectroscopy system (Fabbri et al. 2003)

typical HI-SRS system mainly includes light source, lens, imaging spectrograph, and camera. The detail was described elsewhere (Cen and Lu 2010). An optimized instrument named “Optical Property Analyzer (OPA)” for measuring optical properties was exploited by Lu et al. (2013) based on HI-SRS technique. Combined with suitable inverse algorithms, the main optical properties, such as μ_a , μ_s , and μ_s' , can be obtained from the measured spatially resolved diffuse reflectance profile.

HI-SRS technique can realize a relatively fast measurement on absorption and scattering properties of biological tissues, especially food and agricultural products, over a broad spectral range. Based on obtained spatially resolved diffuse reflectance images, Qin and Lu (2008) determined the absorption and reduced scattering coefficients of some fruit or vegetable samples, such as kiwifruit, apple, pear, plum, peach, tomato, cucumber, and zucchini squash and found that the firmness of tomato was positively correlated with the value of reduced scattering coefficient. Zamora-Rojas et al. (2014) investigated the NIR radiation propagation in pig skin and reported that the dermis layer absorbed most of the light. Besides these, HI-SRS technique has also been used to estimate the maturity of peaches (Cen et al. 2011) and tomato (Huang et al. 2018a; Zhu et al. 2015), to predict the quality of apple (Peng and Lu 2008), peach (Cen et al. 2011; Cen et al. 2012), and tomato (Huang et al., 2018b), to characterize fat content of milk (Qin and Lu 2007), and to predict the tenderness of pork (Sun et al. 2019).

Although SRS techniques are effective means in measuring absorption and scattering properties, more sophisticated instrumentations and more precise mathematical models or algorithms are needed in measurement and calculation (Cen and Lu 2009).

Frequency-Domain Technique

Based on spatially modulated illumination, the frequency-domain resolution technique allows sample being characterized in spatial frequency domain. As shown in Fig. 8, due to absorption and scattering on light by medium, the light detected by the system has different amplitude and phase from those of incident light. In the system of frequency domain, after high frequency modulation (frequency f), the light source incident on the sample, the amplitude change amount M , and the phase retardation angle θ at the radial position r are obtained. Combined with the analytical solution of the diffuse approximation equation, the optical properties can be estimated (Patterson et al. 1991).

Different from the single point illumination of SRS and FD, Cuccia et al. (2005) and Anderson et al. (2007) proposed another spatial frequency-domain imaging (SFDI) system. The typical configuration of the SFDI platform generally consists of an illumination source (e.g., digital projector) and a detector (e.g., CCD camera). Figure 9 shows the schematic

Fig. 7 Hyperspectral imaging-based spatially resolved spectroscopy method: **a** schematic showing the major components of the system and **b** top view of a multi-line-scanning mode for acquiring spatially resolved reflectance profiles (Cen and Lu 2010)

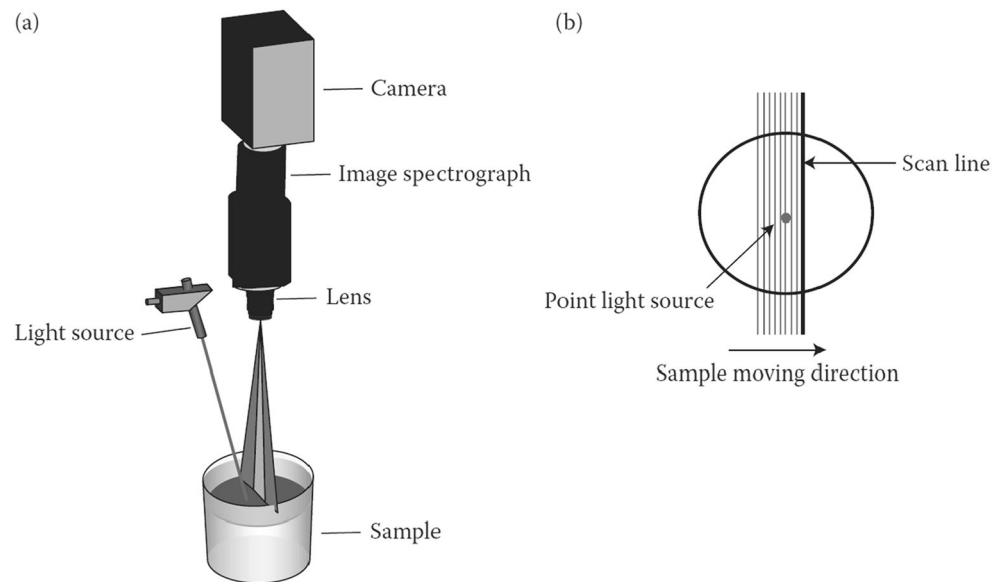


diagram of the SFDI system exploited by Anderson et al. (2007) for multispectral imaging. Based on diffusion theory, μ_a and μ_s' of the measured material can be quantified by using appropriate light propagation model through analyzing the frequency-dependent reflectance.

SFDI has shown great potential in characterizing optical properties of biological materials (Anderson et al. 2007; Hu et al., 2017) and in tissue-simulating model samples (Bodenschatz et al. 2014; Greening et al. 2014; Weber et al. 2011). In addition, it has been successfully applied in the field of medical (Shi et al. 2015), such as early assessment of burn severity in human tissue (Poon et al. 2018).

Time-Resolved Spectroscopy

In the system of time-resolved spectroscopy (TRS), a pulse of monochromatic light that lasts for tens of picoseconds is injected into the diffusive medium by using a suitable optical fiber. The light pulse propagates inside the medium until it

eventually re-emits across the boundary, or it is captured by an absorbing center. The light pulse can be detected at a certain distance (typically in the range from 10 to 40 mm) from the injection point.

The instrument response function, i.e., the injected light pulse, and the broadened distribution of time-of-flight (DTOF) detected at a distance from the injected point are expressed as a function of time, shown in Fig. 10. Since the light propagates inside the diffusive medium, the DTOF of photon is delayed, broadened, and attenuated with respect to the injected light pulse. The delay is caused by limited time that light travels the distance from the injection to detection point. Owing to multiple scattering, different paths traveled by photons are main reason of broadening. Since both absorption and scattering remove photons from considered direction, the attenuation appears.

Because absorption and scattering have different effects on DTOF, μ_a and μ_s' can be estimated utilizing a single-distance time domain measurement for homogeneous media. The difference between TRS and other approaches is that TRS is grounded on the physics of photon migration in diffusive media (Vanoli et al. 2018). With respect to the injection time, the arrival time of photons is proportional to the penetration depth inside the tissue (Del et al. 2002; Steinbrink et al. 2001). The larger the arrival time of photons with respect to the injection time, the deeper the photons penetrate inside the tissue. It was reported that the sensitive depth of light propagation measured by TRS system in apple was about 2 cm, and was not affected by apple skin (Cubeddu et al. 2001b). Moreover, the depth did not depend on source-detector distance.

TRS has been used to estimate absorption and scattering properties of apple (Cubeddu et al. 2001b), to quantize chemical and physical properties of fruit (Cubeddu et al. 2001a;

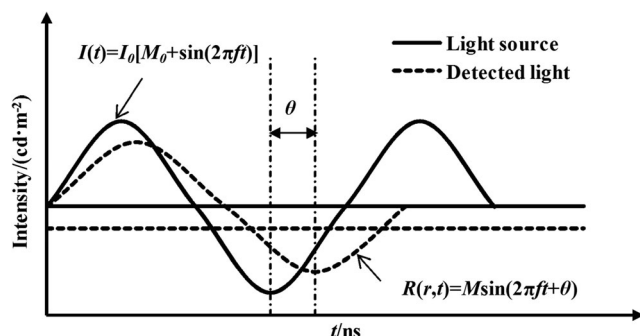
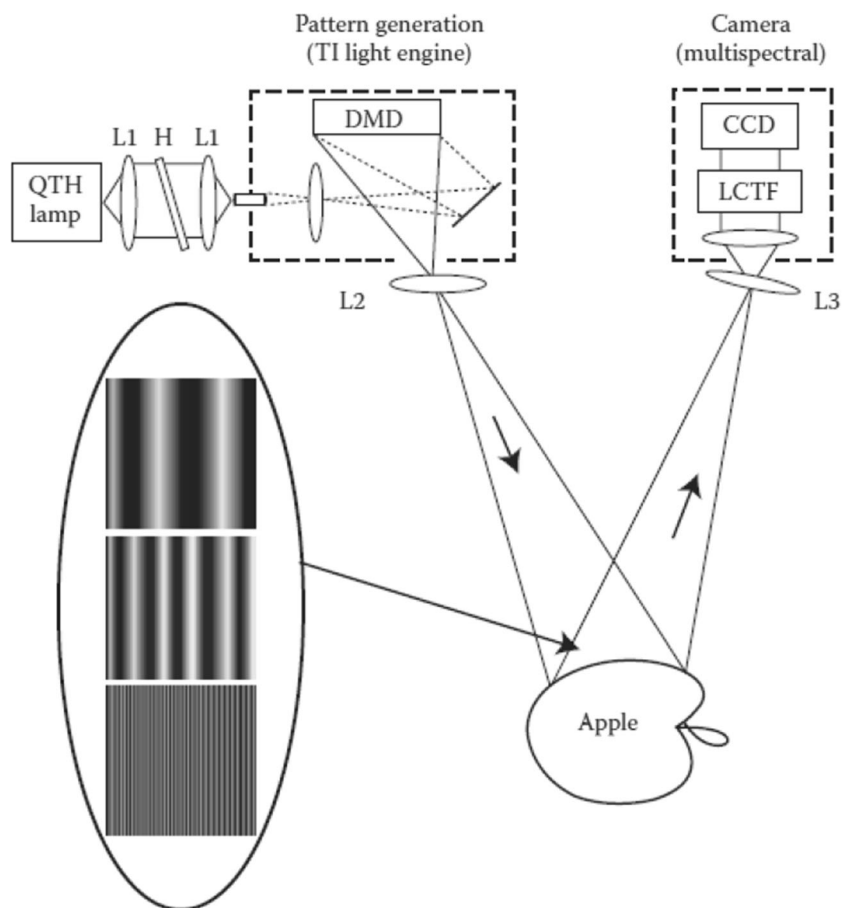


Fig. 8 Schematic diagram of frequency-domain technique (Pan et al. 2018)

Fig. 9 Spatial frequency-domain imaging system: CCD—camera, DMD—digital micromirror device, H—hybridhot mirror, L1—aspheric condenser, L2—projection lens, L3—camera lens, LCTF—liquid crystal-tunable filter, QTH—quartz tungsten halogen (Anderson et al. 2007)



Cubeddu et al. 2001b), and to detect internal quality of kiwi-fruit (Valero et al. 2004) and mealiness of apple (Valero et al. 2005). However, TRS technology is expensive and not commercially available, limiting its further development.

Among the above-described methods, IS is a direct method and is relatively easy to execute, but it is invasive. Simple geometries (generally thin slabs) are required when the transmittance and reflectance of samples are measured using IS. Except for IS, other methods are indirect methods. Contrast with direct method, indirect methods are able to achieve non-destructive or noninvasive acquisition on optical parameters

from intact biological samples, but sophisticated and expensive instruments and complex mathematical models are needed in indirect methods. However, direct method is more challenging than indirect methods due to the difficulty in handling thin samples. Therefore, recent researches have been focused on indirect methods to obtain absorption and scattering properties of food. Table 1 lists the characteristics of different techniques applied in estimating absorption and scattering properties of turbid food.

Up to now, the techniques used to measure the properties of absorption and scattering have been used to study the relationship between optical properties and quality indices, such as sugar content (or soluble solids content), firmness, acidity (or pH), fat content, and lightness, and also been applied to detect maturity or ripeness, bruise, mealiness, juicy of fruit, or vegetable. The application of these measurement techniques on optical properties of turbid food is summarized in Table 2, including the used number of samples, detected quality indicators, and obtained results, such as correlation coefficient or determination coefficient between optical properties and quality indicators, root-mean-squares errors of prediction, and classification accuracy.

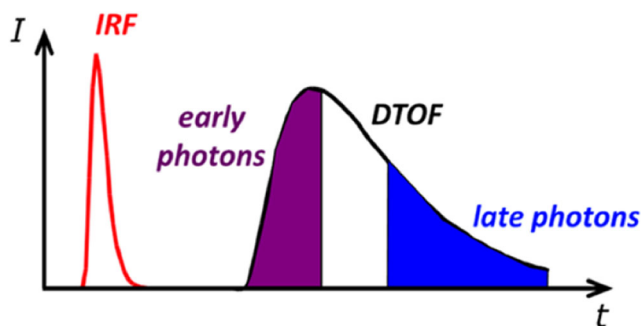


Fig. 10 Schematic diagram of a time-resolved spectroscopy (Pifferi et al. 2016)

Table 1 Characteristics of the techniques applied in measuring optical properties

Measurement techniques		Characteristics
IS	SIS	Low cost, easy to execute, but low speed.
	DIS	Higher speed compared to SIS, but existed error causes uncertainties of measurement.
SRS	Fiber-optic probe	High signal-noise ratio, but more measurement time needed. Fluctuations existed and low practice for solid food.
	MI-SRS	Only valid at single or several wavelengths.
	HI-SRS	Relatively fast measurement, broad spectral range, combined spectral and spatial information.
FD	FD	Cheaper and simpler instruments needed, simple arithmetic operations, but implemented at single frequency.
	SFDI	Multispectral imaging.
TRS		Sensitive to deep structure, independent on source-detector distance, but expensive and not commercially available, and adapt to small but strongly diffusive samples.

DIS, double integrating sphere; *FD*, frequency-domain; *HI-SRS*, hyperspectral imaging-based spatially resolved spectroscopy; *IS*, integrating sphere; *MI-SRS*, monochromatic imaging-based spatially resolved spectroscopy; *SIS*, single integrating sphere; *SRS*, spatially resolved spectroscopy; *SFDI*, spatial frequency-domain imaging; *TRS*, time-resolved spectroscopy

Calculation Algorithms on Absorption and Scattering Properties

Up to now, some methods have been put forward and utilized to calculate the absorption and scattering properties of mediums. Of which, the commonly used methods are IAD, MC, and DA. The basic algorithms of these methods are introduced as follows.

Inverse Adding Doubling

The adding double (AD) method is used to solve the radiative transport equation in a slab geometry for tissue optics. The AD method was extended in an inverse manner, i.e., IAD method, by Prahl et al. (1993) to calculate the absorption and scattering properties of biological materials. Due to much less computational time and high accuracy that are comparable to MC method, IAD method has become one of the most common approaches applied in calculating absorption and scattering properties of biological materials. It has been confirmed to be the accurate and fast method in measuring absorption and scattering properties of thin slab-shaped biological medium. Figure 11 shows the general procedure of IAD algorithm.

As shown in Fig. 11, the first step is measuring reflectance and transmittance of a sample by using IS technique. Then, make a guess on the values of the optical parameters, including μ_a , μ_s' , and g , of the sample. Based on the supposed data, the reflectance and transmittance of the sample are calculated by using AD method. If the difference between the calculated and the measured reflectance and transmittance has a larger error than preset threshold, new data on the optical parameters are supposed, and the reflectance and transmittance are calculated again until the calculated match the measured data well. Then the supposed values in the last time are regarded as the true absorption and scattering properties.

It has been reported that the calculation results based on IAD method are influenced by light loss owing to scattering through glass slides on sample holder and boundary of a sample (Pickering et al. 1993; Prahl 2011). To reduce calculation error of IAD methods caused by light loss, the sample should be a flat slab and completely cover the port of the integrating sphere. Moreover, the diameter of the incident beam should be not only small but also much smaller than the diameters of entrance and exit ports of the integrating sphere.

IAD method combined with IS technique has been utilized to calculate absorption and scattering properties of fruit, such as apple flesh and apple skin (Saeys et al. 2008), kiwifruit tissue (Fang et al., 2016), blueberry (Zhang et al. 2019) and pear (He et al. 2016), vegetable, such as onion flesh and onion skin (Wang and Li 2012), healthy and diseased onion tissues (Wang et al. 2014) and potato (López-Maestresalas et al. 2016), milk (Aernouts et al. 2015a), and meat (Zamora-Rojas et al. 2013). Owing to its high accuracy, IAD has been used as a reference measurement method in developing other techniques, such as MC method (Qin and Lu 2009) and SR method (Cen and Lu 2010) for estimating absorption and scattering properties.

Monte Carlo

MC simulation is an extensive class of computational algorithms that depend on repeated random sampling to obtain numerical results. It is mainly used in numerical integration, optimization, and generating draws from a probability distribution. In optics, MC is valid for a wide range of absorption and scattering coefficients and photon paths. Therefore, it is usually used to simulate light propagation in turbid materials under the assumption of uniform macroscopic optical properties over the ranges of ultraviolet, Vis/NIR light (Palmer and Ramanujam 2006). MC method provides an accurate and flexible mean for

Table 2 Applications of the measurement techniques on optical properties of turbid food

Measurement techniques	Products	Number of samples	Detected quality indicator	Results	References
IS	Apple	42	Skin and flesh	— ^a	(Saeys et al. 2008)
	Apple	155	Bruising degree	Bruising degree classification accuracy = 92.5%.	(Zhang et al. 2017)
	Kiwifruit	30	Three parts of the kiwifruit	—	(Fang, Z.-h., Fu, X.-p., He, X.-m. 2016)
	Pear	130	Firmness and SSC	Firmness: $R = 0.48$; SSC: $R = 0.40$.	(He et al. 2016)
	Blueberry	150	Bruising	—	(Zhang et al. 2019)
	Onion	80	Skin and flesh	—	(Wang and Li 2013)
	Potato	53	Flesh	—	(López-Maestresalas et al. 2016)
	Pig tissue	10	—	—	(Zamora-Rojas et al. 2013)
	Milk	8	Fat globule size	$R^2 \geq 0.990$	(Aernouts et al. 2015a)
	Milk	1200	Milk fat and casein	Fat: $R \geq 0.975$ (1300–1400 nm)	(Aernouts et al. 2015b)
SRS	Fiber-optic probe	Apple	Firmness and SSC	Firmness: $R = 0.844$, RMSEP = 0.987; SSC: $R = 0.901$, RMSEP = 0.686.	(Nghia Nguyen Do et al. 2013)
		Apple	Firmness and SSC	Firmness: $R^2 = 0.71$, RMSEP = 9.68 N; SSC: $R^2 = 0.81$, RMSEP = 0.69%.	(Nghia Nguyen Do et al. 2014a)
		Beef muscle	Tenderness	$R^2 = 0.59$ (between μ_s ' and cooked Warner-Bratzler shear force).	(Xia et al. 2007)
		Beef muscle	—	—	(Xia et al. 2008)
	MI-SRS	Citrus	Early decay detection	Classification accuracy = 96.1%.	(Lorente et al. 2013)
		Banana	Chlorophyll, elasticity, SSC and ripeness	Chlorophyll: $R = 0.9768$ –0.9807; Elasticity: $R = 0.9553$ –0.9759; SSC: $R = 0.9640$ –0.9801; Ripeness: classification accuracy = 97.53%.	(Adebayo et al. 2016)
	HI-SRS	Apple	Firmness and SSC	Firmness: $R = 0.894$, RMSEP = 6.14 N; SSC: $R = 0.883$, RMSEP = 0.73%.	(Peng and Lu 2008)
		Peach	Firmness, SSC, skin lightness and flesh lightness	Firmness: $R = 0.749$, RMSEP = 17.39 N; SSC: $R = 0.504$, RMSEP = 0.92 °Brix; Skin lightness: $R = 0.898$, RMSEP = 3.45; Flesh lightness: $R = 0.741$, RMSEP = 3.27.	(Cen et al. 2012)
		Tomato	Ripeness evaluation	Ripeness: classification accuracy = 88.4%.	(Zhu et al. 2015)
		Tomato	Maturity classification	Maturity: classification accuracy = 81.3–96.3%.	(Huang et al., 2018a)
		Tomato	Firmness, SSC and pH	Firmness: $R = 0.835$; SSC: $R = 0.623$; pH: $R = 0.769$.	(Huang et al. 2018b)
		Pork	Pork tenderness prediction	$R = 0.902$, RMSEP = 5.218 N.	(Sun et al. 2019)
		Milk	Fat content	$R(\mu_{a,600}) = 0.995$; $R(\mu_{s',600}) = 0.998$	(Qin and Lu 2007)
FD	Apple	—	Bruise detection	—	(Anderson et al. 2007)
TRS	Apple, peach, kiwifruit and tomato	—	—	—	(Cubeddu et al. 2001a)
	Apple	30	—	—	(Cubeddu et al. 2001b)
	Apple	90	Mealiness detection	Mealiness identification accuracy: 47–100%.	(Valero et al. 2005)

Table 2 (continued)

Measurement techniques	Products	Number of samples	Detected quality indicator	Results	References
	Apple	150	Juicy and mealy apples	Juicy apple: 57.3% identification rate; Mealy apple: 71.9% identification rate.	(Rizzolo et al. 2010)
	Kiwifruit	180	Firmness, sugar content and acidity	Firmness: classification accuracy = 75%; Sugar content: classification accuracy = 60%; Acidity: classification accuracy = 97%.	(Valero et al. 2004)

IS, integrating sphere; SRS, spatially resolved spectroscopy; MI-SRS, monochromatic imaging-based spatially resolved spectroscopy; HI-SRS, hyperspectral imaging-based spatially resolved spectroscopy; FD, frequency-domain; TRS, time-resolved spectroscopy; SSC, soluble solids content; R, correlation coefficient; R^2 , coefficient of determination; RMSEP, root-mean-squares error of prediction

^a Means that the result was not mentioned in the reference

quantifying light propagation properties that has great challenge to be obtained experimentally.

To obtain an acceptable statistical accuracy, MC method needs a large number of simulations (Alerstam et al. 2008; Fang and Boas 2009). Figure 12 indicates the general steps that an individual photon how to move in MC simulations. Typically, when 10^5 – 10^7 photons are launched, the step size, deflection angle θ , and absorbed fraction are dependent on absorption and scattering properties, which in turn determine the photon position at each step and each time. Unlike diffusion equation, this technique is computationally intensive because a large number of photons have to be simulated to obtain acceptable results. This tells that MC method has a trade-off between computation time and accuracy, i.e., high accuracy could be obtained at more computation time.

To improve the computational speed of MC simulations, hybrid MC, perturbation MC, scaling MC, variance reduction, and parallel computing have been proposed. Since MC has high accuracy and flexible applicability (Boas et al. 2002; Wang et al. 1995), it has been known as the golden standard (Bevilacqua and Depeursinge 1999; Wang et al. 1995). Recently, MC simulations have been used in estimating the absorption and scattering properties of kiwifruit (Baranyai and

Zude 2009), apple (Qin and Lu 2009), and meat (Zamora-Rojas et al. 2014), optimizing HI-SRS measurement system (Cen and Lu 2010) and offering more information about the light distribution inside apple (Fraser 2001) and mandarin fruit (Fraser et al. 2003).

Diffuse Approximation

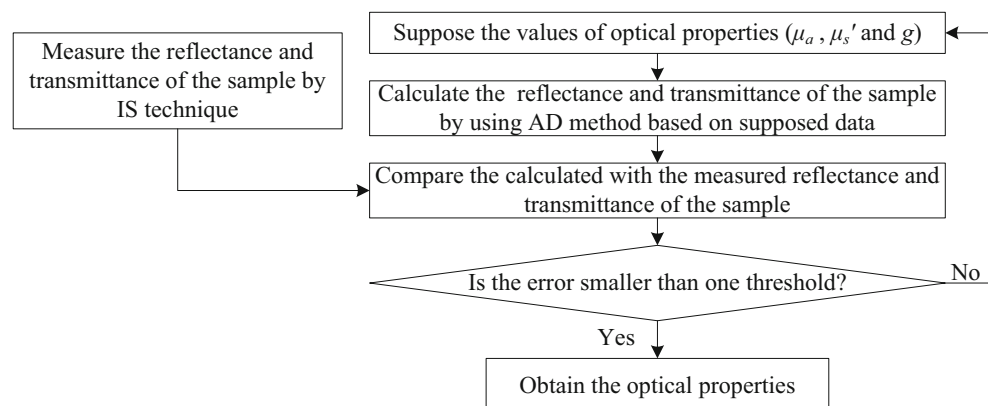
DA is another method for calculating absorption and scattering properties of turbid materials (Alexandrakis et al. 1998; Farrell et al. 1992). DA equation is extracted from radiative transfer equation (RTE) based on several assumptions and simplifications (Star 2010). The DA equation is

$$\frac{1}{c} \frac{\partial \phi(r, t)}{\partial t} = D \nabla^2 \phi(r, t) - \mu_a(r) \phi(r, t) + S_0(r, t) \quad (13)$$

$$D = \frac{1}{3(\mu_a + \mu_s')} \quad (14)$$

where c is the speed of light, t is the time, D is diffusion coefficient (Eq. 14), ∇^2 is the laplaction operator, $\phi(r; t)$ is

Fig. 11 General procedure of inverse adding doubling (IAD) algorithm



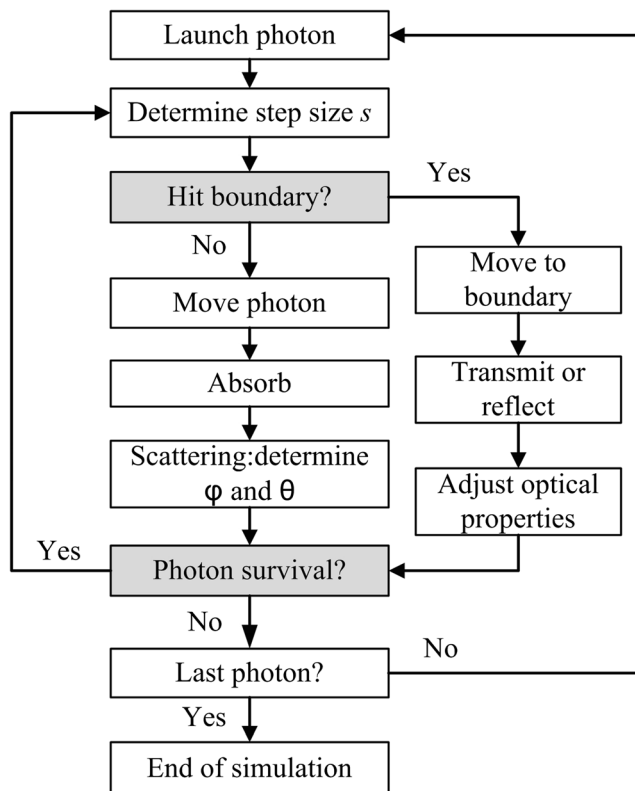


Fig. 12 Generic flowchart on how photon movement is simulated in Monte Carlo algorithm (Watté et al. 2016)

fluence rate in W/m^2 , and $S_0(r,t)$ is isotropic source. More details about the DA equation could be found elsewhere (Welch and Gemert 2011).

DA method is only valid in the case when scattering dominates absorption and the measured spatial locations of diffuse reflected lights are far enough, usually larger than $10 \times (\mu_a + \mu_s')^{-1}$ from illumination point to allow the light to become sufficiently diffuse (Kienle et al. 1996). In many cases, these conditions are not met. Therefore, large errors are usually existed in estimated absorption and scattering properties by using DA. Moreover, DA is not suitable for optically thin samples. Combined with SRS (Kienle et al. 1996), TRS (Vanoli et al. 2014), or FD spectroscopy (Anderson et al. 2007), DA has been applied to calculate absorption and scattering properties of turbid food.

Among the three methods used to calculate absorption and scattering properties of turbid food, IAD method has high accuracy, short computation time, no limits to highly scattering media, but its results are affected by light loss. MC method is a golden standard method, and it has high accuracy and flexible applicability, but long computation time is needed. DA method has strict conditions, causing it has narrow application field.

Generally, the above-described measurement techniques combined with calculation algorithms provide access to

estimate absorption and scattering properties of food and agricultural products.

Conclusions

In this study, some common optical parameters were introduced, the usually used techniques and algorithms in measuring and calculating absorption and scattering properties of turbid food were presented, and the application of these techniques in estimating absorption and scattering properties of turbid food in Vis/NIR range were summarized. Among the measurement techniques, the indirect methods (except for IS) allow noninvasive or nondestructive measurement, but complex mathematical models and sophisticated instrumentation are needed. IS, a direct method, is relatively easy to conduct but is invasive. Further works are required to optimize equipment and simplify model for indirect methods and to overcome the difficulty in handling thin samples for IS.

As for the calculation algorithms, IAD method has been confirmed to be accurate and fast, but it is easily affected by light loss in estimating absorption and scattering properties. Although MC algorithm has high accuracy, it needs long computation time. DA is valid only when several assumptions and simplifications meet. Optimizing MC and DA methods and reducing light loss as far as possible in IAD are what should be focused on in the future.

Funding Information This study received financial support from the National Natural Science Foundation of China (No. 31772065).

References

- Adebayo, S. E., Hashim, N., Abdan, K., Hanafi, M., & Mollazade, K. (2016). Prediction of quality attributes and ripeness classification of bananas using optical properties. *Scientia Horticulturae*, 212, 171–182.
- Aernouts, B., Van Beers, R., Watte, R., Huybrechts, T., Jordens, J., Vermeulen, D., Van Gerven, T., Lammertyn, J., & Saeys, W. (2015a). Effect of ultrasonic homogenization on the Vis/NIR bulk optical properties of milk. *Colloids and Surfaces. B, Biointerfaces*, 126, 510–519.
- Aernouts, B., Van Beers, R., Watte, R., Huybrechts, T., Lammertyn, J., & Saeys, W. (2015b). Visible and near-infrared bulk optical properties of raw milk. *Journal of Dairy Science*, 98, 6727–6738.
- Alerstam, E., Svensson, T., & Andersson-Engels, S. (2008). Parallel computing with graphics processing units for high-speed Monte Carlo simulation of photon migration. *Journal of Biomedical Optics*, 13(6), 060504.
- Alexandrakis, G., Farrell, T. J., & Patterson, M. S. (1998). Accuracy of the diffusion approximation in determining the optical properties of a two-layer turbid medium. *Applied Optics*, 37(31), 7401–7409.
- Anderson, E.R., Cuccia, D.J., Durkin, A.J., (2007). Detection of bruises on golden delicious apples using spatial-frequency-domain imaging, in: Tuan Vo-Dinh, W.S.G, David A. Benaron, Gerald E. Cohn, Ramesh Raghavachari (Eds.), *Advanced Biomedical and Clinical Diagnostic Systems V*. Processing of SPIE, San Jose,CA,USA.

- Baranyai, L., & Zude, M. (2009). Analysis of laser light propagation in kiwifruit using backscattering imaging and Monte Carlo simulation. *Computers and Electronics in Agriculture*, 69(1), 33–39.
- Bevilacqua, F., & Depeursinge, C. (1999). Monte Carlo study of diffuse reflectance at source–detector separations close to one transport mean free path. *Journal of the Optical Society of America*, 16(12), 2935–2945.
- Boas, D. A., Culver, J. P., Stott, J. J., & Dunn, A. K. (2002). Three dimensional Monte Carlo code for photon migration through complex heterogeneous media including the adult human head. *Optics Express*, 10(3), 159–170.
- Bodenschatz, N., Brandes, A., Liemert, A., & Kienle, A. (2014). Sources of errors in spatial frequency domain imaging of scattering media. *Journal of Biomedical Optics*, 19(7), 071405.
- Carp, S. A., Farzam, P., Redes, N., Hueber, D. M., & Franceschini, M. A. (2017). Combined multi-distance frequency domain and diffuse correlation spectroscopy system with simultaneous data acquisition and real-time analysis. *Biomedical Optics Express*, 8(9), 3993–4006.
- Cen, H., & Lu, R. (2009). Quantification of the optical properties of two-layer turbid materials using a hyperspectral imaging-based spatially-resolved technique. *Applied Optics*, 48(29), 5612–5623.
- Cen, H., & Lu, R. (2010). Optimization of the hyperspectral imaging-based spatially-resolved system for measuring the optical properties of biological materials. *Optics Express*, 18(16), 17412–17432.
- Cen, H., Lu, R., Mendoza, F. A., & Ariana, D. P. (2011). Peach maturity/quality assessment using hyperspectral imaging-based spatially-resolved technique. In M. S. Kim, S. I. Tu, & K. Chao (Eds.), *Sensing for agriculture and food quality and safety II*. Orlando: Processing of SPIE.
- Cen, H., Lu, R., Mendoza, F. A., & Ariana, D. P. (2012). Assessing multiple quality attributes of peaches using optical absorption and scattering properties. *Transactions of the ASABE*, 55(2), 647–657.
- Cen, H., Lu, R., Nguyen-Do, N., & Saeys, W. (2016). Spatially resolved spectroscopic technique for measuring optical properties of food. In R. Lu (Ed.), *Light scattering technology for food property, quality and safety assessment*. Boca Raton: Taylor & Francis Group, CRC Press.
- Cubeddu, R., D'Andrea, C., Pifferi, A., Taroni, P., Torricelli, A., Valentini, G., Dover, C., Johnson, D., Ruizaltisent, M., & Valero, C. (2001a). Nondestructive quantification of chemical and physical properties of fruits by time-resolved reflectance spectroscopy in the wavelength range 650–1000 nm. *Applied Optics*, 40(4), 538–543.
- Cubeddu, R., D'Andrea, C., Pifferi, A., Taroni, P., Torricelli, A., Valentini, G., Ruizaltisent, M., Valero, C., Ortiz, C., & Dover, C. (2001b). Time-resolved reflectance spectroscopy applied to the nondestructive monitoring of the internal optical properties in apples. *Applied Spectroscopy*, 55(10), 1368–1374.
- Cuccia, D. J., Bevilacqua, F., Durkin, A. J., & Tromberg, B. J. (2005). Modulated imaging: quantitative analysis and tomography of turbid media in the spatial-frequency domain. *Optics Letters*, 30(11), 1354–1356.
- Del, B. S., Martelli, F., & Zaccanti, G. (2002). Penetration depth of light re-emitted by a diffusive medium: theoretical and experimental investigation. *Physics in Medicine & Biology*, 47(23), 4131–4144.
- Fabbri, F., Franceschini, M. A., & Fantini, S. (2003). Characterization of spatial and temporal variations in the optical properties of tissuelike media with diffuse reflectance imaging. *Applied Optics*, 42(16), 3063–3072.
- Fang, Q., & Boas, D. A. (2009). Monte Carlo simulation of photon migration in 3D turbid media accelerated by graphics processing units. *Optics Express*, 17(22), 20178–20190.
- Fang, Z. H., Fu, X. P., & He, X. M. (2016). Investigation of absorption and scattering characteristics of kiwifruit tissue using a single integrating sphere system. *Journal of Zhejiang University-Science B(Biomedicine & Biotechnology)*, 17(6), 484–492 (in Chinese).
- Farrell, T. J., Patterson, M. S., & Wilson, B. (1992). A diffusion-theory model of spatially resolved, steady-state diffuse reflectance for the noninvasive determination of tissue optical-properties in vivo. *Medical Physics*, 19(4), 879–888.
- Fraser, D. G. (2001). Near-infrared (NIR) light penetration into an apple. *Postharvest Biology and Technology*, 22(1), 191–195.
- Fraser, D. G., Jordan, R. B., Künemeyer, R., & Mcglone, V. A. (2003). Light distribution inside mandarin fruit during internal quality assessment by NIR spectroscopy. *Postharvest Biology and Technology*, 27(2), 185–196.
- Greening, G. J., Istfan, R., Higgins, L. M., Balachandran, K., Roblyer, D., Pierce, M. C., & Muldoon, T. J. (2014). Characterization of thin poly(dimethylsiloxane)-based tissue-simulating phantoms with tunable reduced scattering and absorption coefficients at visible and near-infrared wavelengths. *Journal of Biomedical Optics*, 19(11).
- He, X., Fu, X., Rao, X., & Fang, Z. (2016). Assessing firmness and SSC of pears based on absorption and scattering properties using an automatic integrating sphere system from 400 to 1150nm. *Postharvest Biology and Technology*, 121, 62–70.
- Herremans, E., Bongaers, E., Estrade, P., Gondek, E., Hertog, M., Jakubczyk, E., Nghia Nguyen Do, T., Rizzolo, A., Saeys, W., Spinelli, L., Torricelli, A., Vanoli, M., Verboven, P., & Nicolai, B. (2013). Microstructure-texture relationships of aerated sugar gels: novel measurement techniques for analysis and control. *Innovative Food Science & Emerging Technologies*, 18(2), 202–211.
- Hu, D., Lu, R., & Ying, Y. (2018). A two-step parameter optimization algorithm for improving estimation of optical properties using spatial frequency domain imaging. *Journal of Quantitative Spectroscopy & Radiative Transfer*, 207, 32–40.
- Hu, W. Y., Wang, Y. H., Zhang, X. S., Li, J. H., Wang, Z. Y., & Zhao, L. L. (2017). Determination and 3D reconstruction of optical parameters of biological tissue based on frequency-domain near-infrared spectroscopy. *Spectroscopy and Spectral Analysis*, 37(5), 1374–1379 (in Chinese).
- Huang, Y., Lu, R., Hu, D., & Chen, K. (2018a). Quality assessment of tomato fruit by optical absorption and scattering properties. *Postharvest Biology and Technology*, 143, 78–85.
- Huang, Y. P., Renfu, L., Qi, C., & Chen, K. J. (2018b). Tomato maturity classification based on spatially resolved spectra. *Spectroscopy and Spectral Analysis*, 38(7), 2183–2188 (in Chinese).
- Jacquez, J. A., & Kuppenheim, H. F. (1955). Theory of the integrating sphere. *Journal of the Optical Society of America*, 45(6), 460–470.
- Kienle, A., Lilge, L., Patterson, M. S., Hibst, R., Steiner, R., & Wilson, B. C. (1996). Spatially resolved absolute diffuse reflectance measurements for noninvasive determination of the optical scattering and absorption coefficients of biological tissue. *Applied Optics*, 35(13), 2304–2314.
- Liu, D., Guo, W., Li, Q., & Xie, D. (2019a). Effect of a plant growth regulator on optical properties of kiwifruit during growth in wavelength range of 950–1650 nm. *Postharvest Biology and Technology*, 156:110918.
- Liu, D., Guo, W., Li, Q., & Xie, D. (2019b). Relationship of the bulk optical properties in 950–1650 nm wavelength range with internal quality and microstructure of kiwifruit during maturation. *Biosystems Engineering*, 84, 45–54.
- López-Maestresalas, A., Aernouts, B., Beers, R. V., Arazuri, S., Jarén, C., Baerdemaeker, J. D., & Saeys, W. (2016). Bulk optical properties of potato flesh in the 500–1900nm range. *Food and Bioprocess Technology*, 9(3), 463–470.
- Lorente, D., Zude, M., Regen, C., Palou, L., Gomez-Sanchis, J., & Blasco, J. (2013). Early decay detection in citrus fruit using laser-light backscattering imaging. *Postharvest Biology and Technology*, 86, 424–430.
- Lu, R. (2016). Overview of light interaction with food and biological materials. In R. Lu (Ed.), *Light scattering Technology for Food*

- Property, quality and safety assessment*. Boca Raton: Taylor & Francis Group, CRC Press.
- Lu, R., Cen, H., & Kilcast, D. (2013). *Non-destructive methods for food texture assessment*. Cambridge: Woodhead Publishing Sawston.
- Nghia Nguyen Do, T., Erkinbaev, C., Nicolai, B., Saeys, W., Tsuta, M., & De Baerdemaeker, J. (2013). Spatially resolved spectroscopy for nondestructive quality measurements of Braeburn apples cultivated in sub-fertilization condition. In N. Kondo (Ed.), *Sensing Technologies for Biomaterial, food, and agriculture 2013*. Orlando: Processing of SPIE.
- Nghia Nguyen Do, T., Erkinbaev, C., Tsuta, M., De Baerdemaeker, J., Nicolai, B., & Saeys, W. (2014a). Spatially resolved diffuse reflectance in the visible and near-infrared wavelength range for non-destructive quality assessment of 'Braeburn' apples. *Postharvest Biology and Technology*, 91, 39–48.
- Nghia Nguyen Do, T., Rizzolo, A., Herremans, E., Vanoli, M., Cortellino, G., Erkinbaev, C., Tsuta, M., Spinelli, L., Contini, D., Torricelli, A., Verboven, P., De Baerdemaeker, J., Nicolai, B., & Saeys, W. (2014b). Optical properties-microstructure-texture relationships of dried apple slices: Spatially resolved diffuse reflectance spectroscopy as a novel technique for analysis and process control. *Innovative Food Science & Emerging Technologies*, 21, 160–168.
- Palmer, G. M., & Ramanujam, N. (2006). Monte Carlo-based inverse model for calculating tissue optical properties. Part I: theory and validation on synthetic phantoms. *Applied Optics*, 45(5), 1062–1071.
- Pan, L., Wen, K., Cao, N., Sun, K., Liu, Q., Tu, K., & Zhu, Q. (2018). Measurement of optical parameters of fruits and vegetables and its application in quality detection. *Journal of Nanjing Agricultural University*, 41(1), 26–37 (in Chinese).
- Patterson, M. S., Chance, B., & Wilson, B. C. (1989). Time resolved reflectance and transmittance for the non-invasive measurement of tissue optical properties. *Applied Optics*, 28(12), 2331–2336.
- Patterson, M. S., Moulton, J. D., & Wilson, B. C. (1991). Frequency-domain reflectance for the determination of the scattering and absorption properties of tissue. *Applied Optics*, 30(31), 4474–4476.
- Peng, Y., & Lu, R. (2008). Analysis of spatially resolved hyperspectral scattering images for assessing apple fruit firmness and soluble solids content. *Postharvest Biology and Technology*, 48(1), 52–62.
- Pickering, J. W., Prah, S. A., Van, W. N., Beek, J. F., Sterenborg, H. J., & van Gemert, M. J. (1993). Double-integrating-sphere system for measuring the optical properties of tissue. *Applied Optics*, 32(4), 399.
- Pifferi, A., Contini, D., Dalla Mora, A., Farina, A., Spinelli, L., & Torricelli, A. (2016). New frontiers in time-domain diffuse optics, a review. *Journal of Biomedical Optics*, 21(9).
- Poon, C., Sunar, U., Rohrbach, D. J., Krishnamurthy, S., Olsen, T., Kent, M., Weir, N. M., Simman, R., & Travers, J. B. (2018). Early assessment of burn severity in human tissue ex vivo with multi-wavelength spatial frequency domain imaging. *Toxicology In Vitro*, 52, 251–254.
- Prah, S. (2011). *Everything I think you should know about inverse adding-doubling*. Orlando: Wilsonville.
- Prah, S. A., van Gemert, M. J., & Welch, A. J. (1993). Determining the optical properties of turbid media by using the adding-doubling method. *Applied Optics*, 32(4), 559–568.
- Qin, J., & Lu, R. (2007). Measurement of the absorption and scattering properties of turbid liquid foods using hyperspectral imaging. *Applied Spectroscopy*, 61(4), 388–396.
- Qin, J., & Lu, R. (2008). Measurement of the optical properties of fruits and vegetables using spatially resolved hyperspectral diffuse reflectance imaging technique. *Postharvest Biology and Technology*, 49(3), 355–365.
- Qin, J. W., & Lu, R. F. (2009). Monte Carlo simulation for quantification of light transport features in apples. *Computers and Electronics in Agriculture*, 68(1), 44–51.
- Rizzolo, A., Vanoli, M., Spinelli, L., & Torricelli, A. (2010). Sensory characteristics, quality and optical properties measured by time-resolved reflectance spectroscopy in stored apples. *Postharvest Biology and Technology*, 58, 1–12.
- Saeys, W., Velazco-Roa, M. A., Thennadil, S. N., Ramon, H., & Nicolai, B. M. (2008). Optical properties of apple skin and flesh in the wavelength range from 350 to 2200 nm. *Applied Optics*, 47(7), 908–919.
- Shi, W., Gao, W., & Chen, C. (2015). Handheld swept source optical coherence tomography for imaging human skin in vivo. *Acta Optica Sinica*, 35(11), 242–250 (in Chinese).
- Star, W. M. (2010). *Diffusion theory of light transport*. New York: Springer Netherlands.
- Steinbrink, J., Wabnitz, H., Obrig, H., Villringer, A., & Rinneberg, H. (2001). Determining changes in NIR absorption using a layered model of the human head. *Physics in Medicine & Biology*, 46(3), 879–896.
- Sun, H., Peng, Y., Zheng, X., Wang, W., & Zhang, J. (2019). Comparative analysis of pork tenderness prediction using different optical scattering parameters. *Journal of Food Engineering*, 248, 1–8.
- Tuchin, V. (2007). Tissue optics: light scattering methods and instruments for medical diagnosis, second edition. *SPIE Optical Engineering*, 13, 882.
- Valero, C., Ruizaltisent, M., Cubeddu, R., Pifferi, A., Taroni, P., Torricelli, A., Valentini, G., Johnson, D. S., & Dover, C. J. (2004). Detection of internal quality in kiwi with time-domain diffuse reflectance spectroscopy. *Applied Engineering in Agriculture*, 20(2), 223–232.
- Valero, C., Barreiro, P., Ruiz-Altisent, M., Cubeddu, R., Pifferi, A., Taroni, P., Torricelli, A., Valentini, G., Johnson, D., & Dover, C. (2005). Mealiness detection in apples using time resolved reflectance spectroscopy. *Journal of Texture Studies*, 36(4), 439–458.
- Vanoli, M., Rizzolo, A., Grassi, M., Spinelli, L., Verlinden, B. E., & Torricelli, A. (2014). Studies on classification models to discriminate 'Braeburn' apples affected by internal browning using the optical properties measured by time-resolved reflectance spectroscopy. *Postharvest Biology and Technology*, 91, 112–121.
- Vanoli, M., Grassi, M., Spinelli, L., Torricelli, A., & Rizzolo, A. (2018). Quality and nutraceutical properties of mango fruit: influence of cultivar and biological age assessed by time-resolved reflectance spectroscopy. *Advances in Horticultural Science*, 32(3), 407–420.
- Verhoelst, E., Bamelis, F., De Ketelaere, B., Nghia Nguyen Do, T., De Baerdemaeker, J., Saeys, W., Tsuta, M., & Decuyper, E. (2011). The potential of spatially resolved spectroscopy for monitoring angiogenesis in the chorioallantoic membrane. *Biotechnology Progress*, 27(6), 1785–1792.
- Wang, L., Jacques, S. L., & Zheng, L. (1995). MCML-Monte Carlo modeling of light transport in multi-layered tissues. *Computer Methods & Programs In Biomedicine*, 47(2), 131–146.
- Wang, W., & Li, C. (2012). The optical properties of onion dry skin and flesh at the wavelength 632.8 nm. In M. S. Kim, S. I. Tu, & K. Chao (Eds.), *Sensing for agriculture and food quality and safety iv*. Orlando: Processing of SPIE.
- Wang, W., & Li, C. (2013). Measurement of the light absorption and scattering properties of onion skin and flesh at 633 nm. *Postharvest Biology and Technology*, 86, 494–501.
- Wang, W., Li, C., & Gitaitis, R. D. (2014). Optical properties of healthy and diseased onion tissues in the visible and near-infrared spectral region. *Transactions of the ASABE*, 57(6), 1771–1782.
- Watte, R., Nghia Nguyen Do, T., Aernouts, B., Erkinbaev, C., De Baerdemaeker, J., Nicolai, B., & Saeys, W. (2013). Metamodeling approach for efficient estimation of optical properties of turbid media from spatially resolved diffuse reflectance measurements. *Optics Express*, 21(26), 32630–32642.
- Watté, R., Aernouts, B., & Saeys, W. (2016). Monte Carlo modeling of light transfer in food. In R. Lu (Ed.), *Light scattering technology for food property, quality and safety assessment*. CRC Press, Boca Raton, FL: Taylor & Francis Group.

- Weber, J. R., Cuccia, D. J., Johnson, W. R., Bearman, G. H., Durkin, A. J., Hsu, M., Lin, A., Binder, D. K., Wilson, D., & Tromberg, B. J. (2011). Multispectral imaging of tissue absorption and scattering using spatial frequency domain imaging and a computed-tomography imaging spectrometer. *Journal of Biomedical Optics*, 16(1), 011015.
- Wei, K., Ma, C., Sun, K., Liu, Q., Zhao, N., Sun, Y., Tu, K., & Pan, L., (2020). Relationship between optical properties and soluble sugar contents of apple flesh during storage. *Postharvest Biology and Technology* 159:111021
- Welch, A. J., & Gemert, M. J. C. V. (2011). *Optical-thermal response of laser-irradiated tissue*. New York: Springer Netherlands.
- Xia, J., Weaver, A., Gerrard, D. E., & Yao, G. (2008). Heating induced optical property changes in beef muscle. *Journal of Food Engineering*, 84, 75–81.
- Xia, J. J., Berg, E. P., Lee, J. W., & Yao, G. (2007). Characterizing beef muscles with optical scattering and absorption coefficients in VIS-NIR region. *Meat Science*, 75(1), 78–83.
- Zamora-Rojas, E., Aernouts, B., Garrido-Varo, A., Perez-Marin, D., Emilio Guerrero-Ginel, J., & Saeys, W. (2013). Double integrating sphere measurements for estimating optical properties of pig subcutaneous adipose tissue. *Innovative Food Science & Emerging Technologies*, 19, 218–226.
- Zamora-Rojas, E., Garrido-Varo, A., Aernouts, B., Pérez-Marín, D., Saeys, W., Yamada, Y., & Guerrero-Ginel, J. E. (2014). Understanding near infrared radiation propagation in pig skin reflectance measurements. *Innovative Food Science & Emerging Technologies*, 22(1), 137–146.
- Zhang, M., Li, C., & Yang, F. (2019). Optical properties of blueberry flesh and skin and Monte Carlo multi-layered simulation of light interaction with fruit tissues. *Postharvest Biology and Technology*, 150, 28–41.
- Zhang, S., Wu, X., Zhang, S., Cheng, Q., & Tan, Z. (2017). An effective method to inspect and classify the bruising degree of apples based on the optical properties. *Postharvest Biology and Technology*, 127, 44–52.
- Zhu, Q., He, C., Lu, R., Mendoza, F., & Cen, H. (2015). Ripeness evaluation of ‘Sun bright’ tomato using optical absorption and scattering properties. *Postharvest Biology and Technology*, 103, 27–34.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.